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JOURNEY FROM AI COLLABORATION TO CODING

FROM PHYSICAL ARCHITECTURE TO DIGITAL ARCHITECTURE

2022-2025

PHD RESEARCH

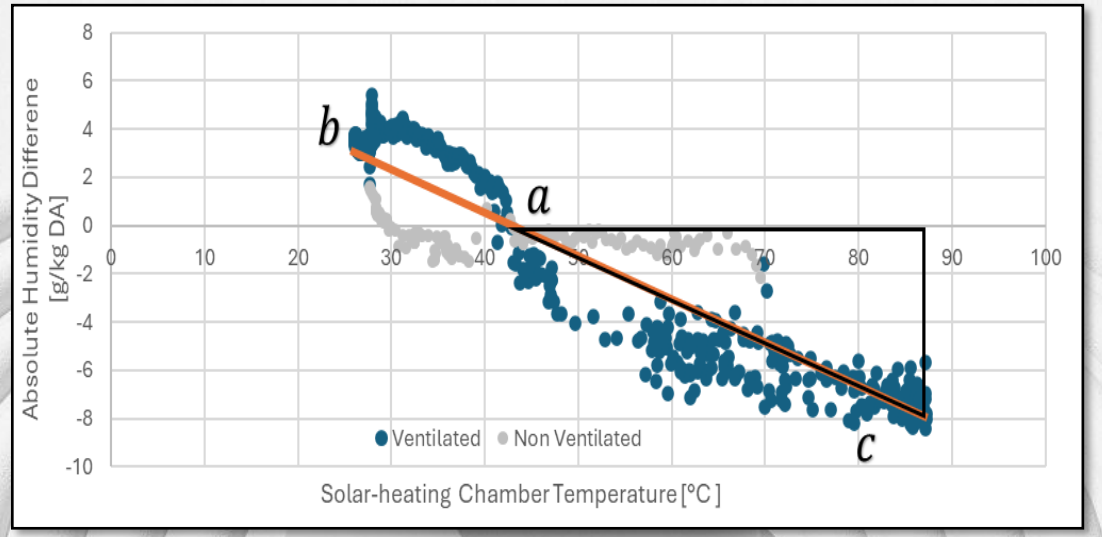
Working with LLMs

Adapting to change

Since 2022, I have integrated Artificial Intelligence into the core of my PhD-level research at Tohoku University. My journey began with utilizing early LLMs (from Microsoft Bing to ChatGPT TO Gemini) as sophisticated research engines.

Verifying AI Information

my primary focus was on Verification and identifying the boundaries between AI-generated hallucinations and verified scientific truth. This period was essential for testing the limits of AI and human knowledge while maintaining the rigorous accuracy required in world class research.



$$\Delta X = C_{sys}.PV_{sat}.(T_{avg} - T_s).\eta_{limit}$$

October 2025

MOMOTARO

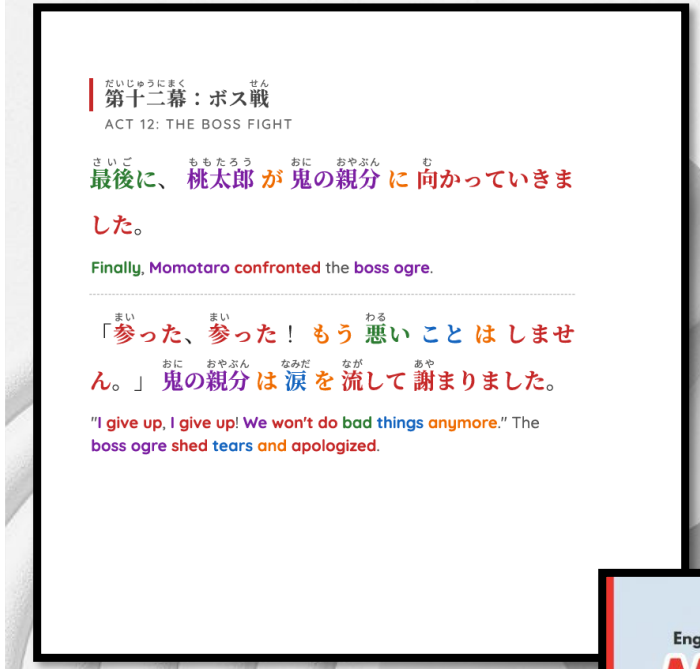
Generative AI

HTML Writing

Initial transition from graphic design into coding. I found that standard word-processing could not support the complex, dual-language formatting I envisioned. Driven by the need for a specific UX (User Experience), I made the decision to build the book's framework using HTML. This allowed for a level of structural precision and layout flexibility that traditional design tools simply could not provide.

Asset Consistency

I also utilized this project as a research ground for generative AI in asset production. I focused on generating high-fidelity, consistent graphic assets—ensuring that character designs, art styles, and world-building elements remained uniform across the entire narrative. This project also served as an end-to-end lesson in global product distribution, as I navigated the complexities of Amazon Self-Publishing to make the book available to a worldwide audience.



December 2025

GAME MAKING

Foundations of Logic loop and Design

Programming and coding

I developed *Demon Hunter* as my first deep dive into functional programming and game logic. The motivation was to explore how complex interactions—such as combat systems, state management, and asset rendering—could be executed within the constraints of a single-file HTML environment. This project served as a training ground for understanding event loops and real-time state updates.

Manual Code adjustment

A key part of this process involved the manual refinement of the codebase to achieve the precise feel and behaviour I envisioned. There were many instances where AI tools could not fully grasp the nuance of my intention, requiring me to step in and manually edit the code to resolve complex logic or fine-tune the user experience.

December 2025

AI SAFEGUARD

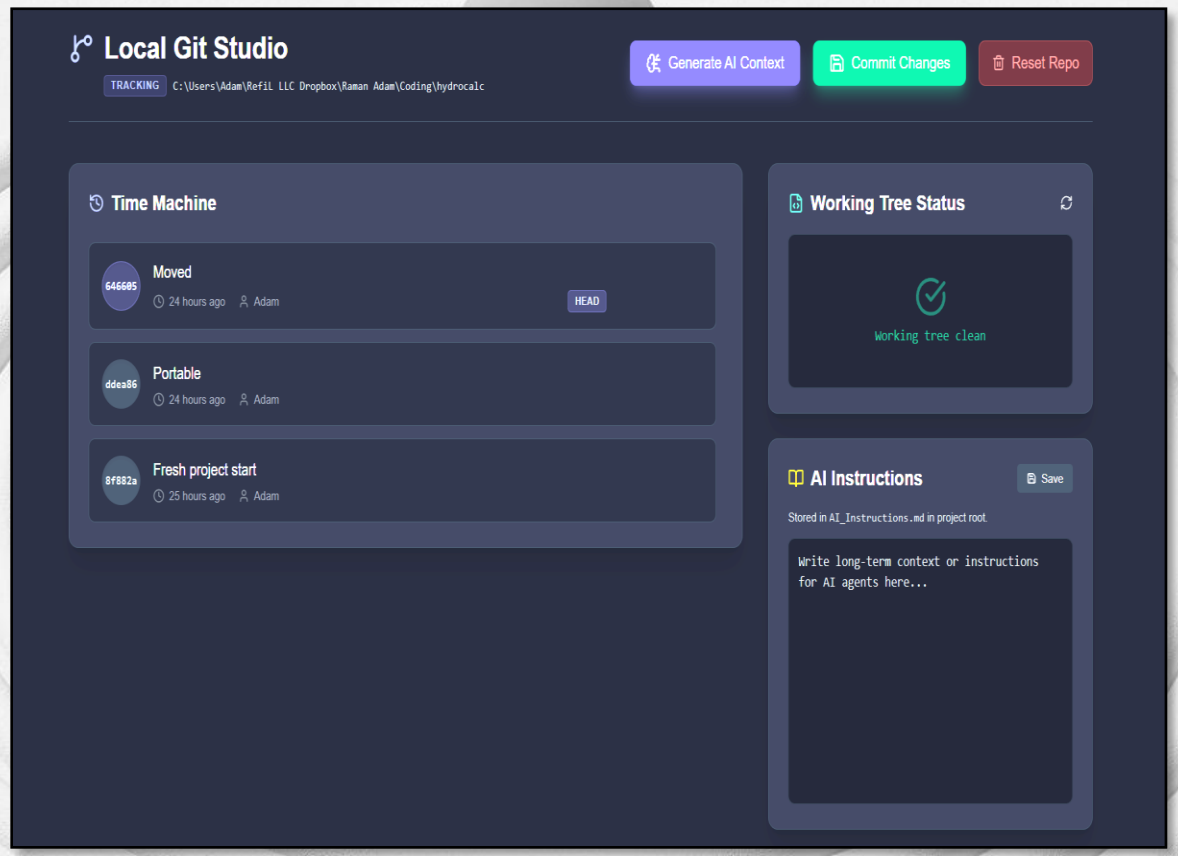
Local Git Studio

Code save points

As I moved toward using AI agents for complex tasks, I realized that AI may break the system. I built a lightweight system that acts as a bridge between my local filesystem and AI agents. The core of this project is the "Time Machine" interface, which allows me to visualize code evolution and restore snapshots instantly, ensuring a safe environment for rapid AI-assisted iteration.

AI CONTEXT AWARENESS

A feature of this system is the **AI Neural Context** system. Instead of constant copy-pasting, I implemented a one-click generation tool that provides AI agents with an overview of the entire codebase. By combining this with persistent project instructions, I created a workflow where the AI never loses track of the overarching goals. This project is a management too for high-speed, agentic development, allowing me to maintain full control while the AI handles the heavy lifting of code generation.



January 2026

DIGITAL LIFE

AGILE DEVELOPMENT

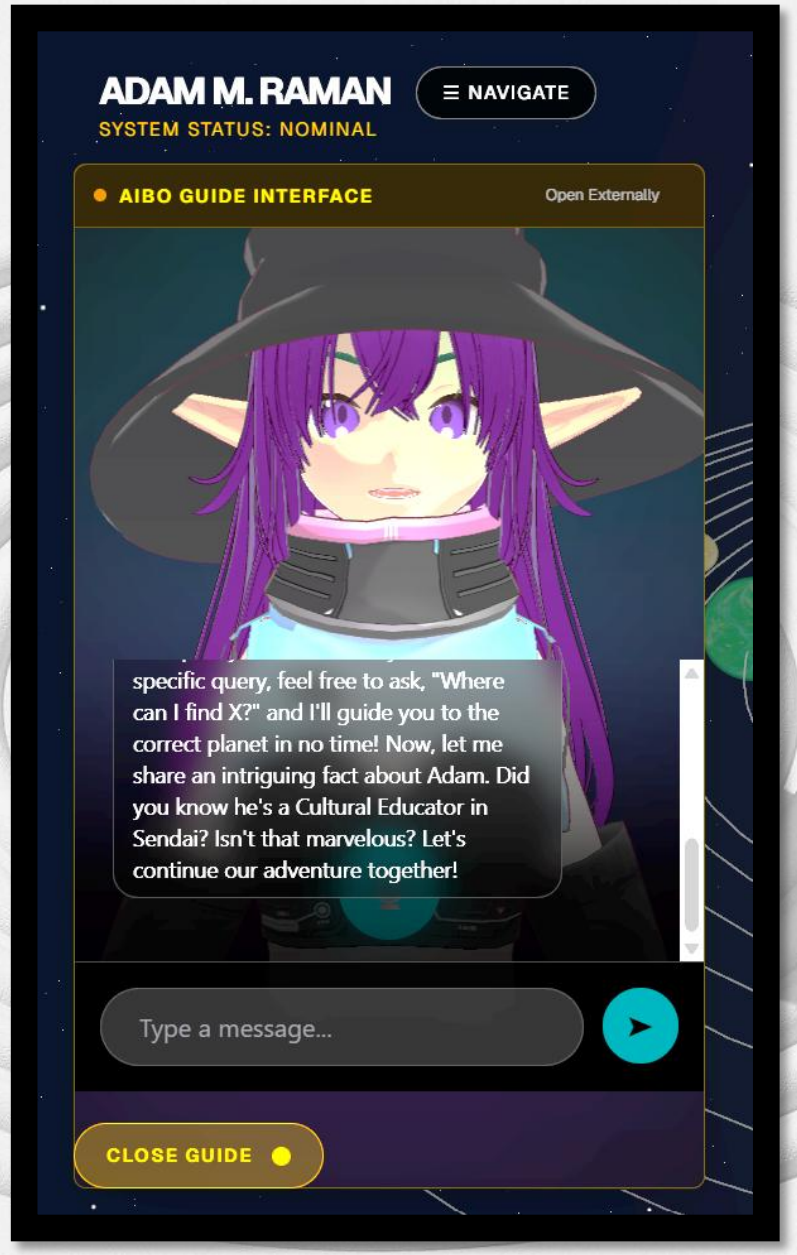
THINK, MOVE, REMEMBER, FEEL

I wanted to create a digital humanoid pet with a true sense of presence. To achieve this, I developed a Multi-Core Architecture system:

The Cognitive Core manages a "Dream Cycle" to consolidate daily interactions into long-term Memory and a "Pulse" system for spontaneous thought.
The Affective Core translates mood into vocal pitch and movement speed.
The Motor Core uses procedural bone manipulation instead of baked animations.

AI Receptionist

While *AIBO_Alive* serves as the complex research goal, *Project-AIBO* is the first step in an agile development series designed for deployment. This version focuses on a "Generative Interface" that translate AI intent into immediate facial expressions. By integrating with **OpenRouter** and **Edge-TTS**, I created a seamless loop where the AI's personality is expressed through synchronized speech and body language. It currently serves as the intelligent receptionist for my *Solar Punk* portfolio.



February 2026

TRUSTED INFO

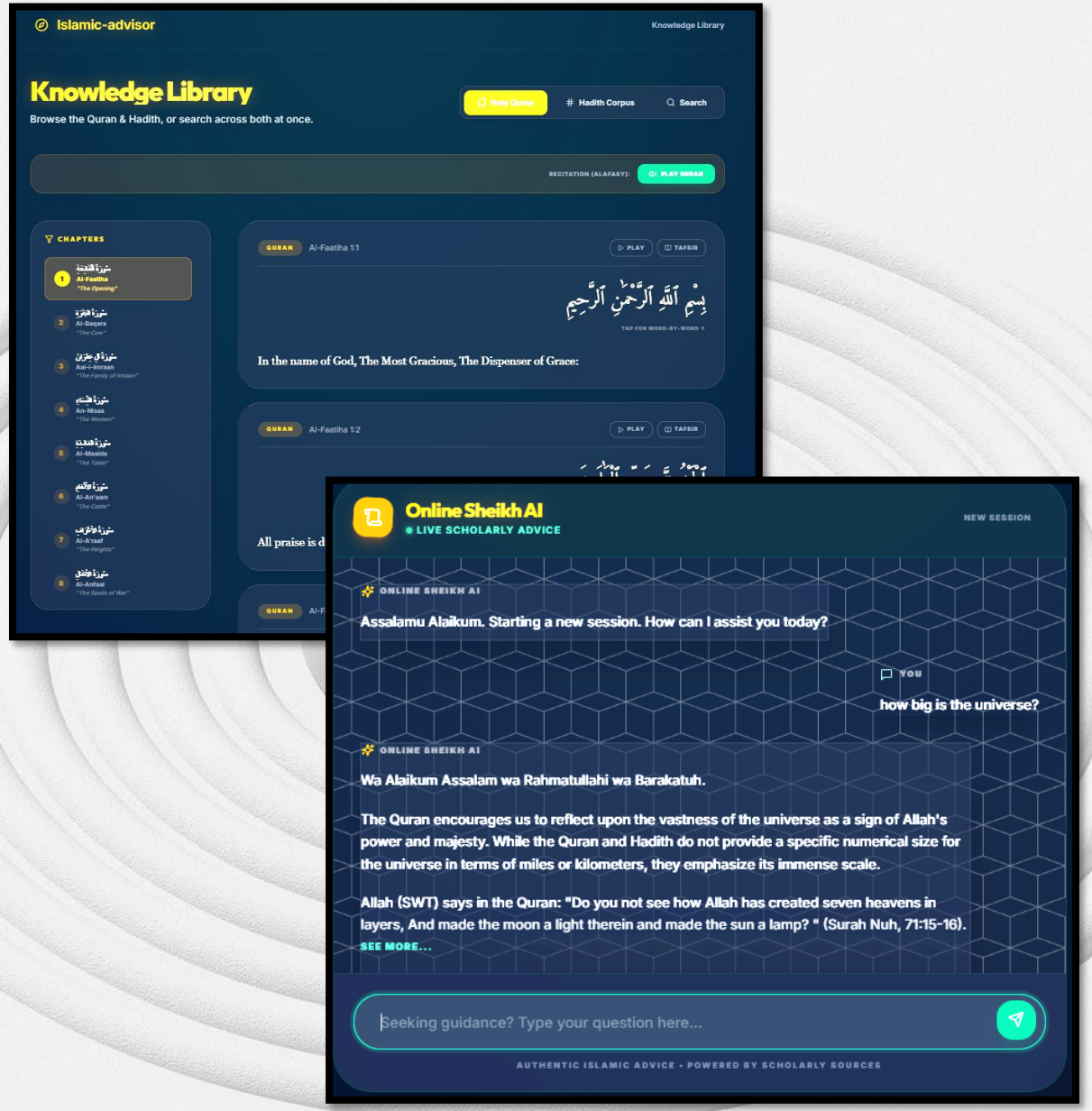
Retrieval-Augmented Generation (RAG)

CITE YOUR SOURCES

I developed the *Islamic Advisor* as a research into AI information integrity. To move away from the "black box" nature of standard LLMs, I created a system that prioritizes verified information over generative guessing. To achieve this, I implemented **RAG** workflow that forces the AI to ground its responses in a **Knowledge Base API**. By connecting the model to trusted, scholarly API sources, I ensured that every answer is backed by documented evidence rather than internal probability.

DECISION MAKING PROCESS

I developed a Decision-making process that occurs before a user ever sees a response:
Query Analysis: The system first identifies the specific intent of the user's question.
Context Retrieval: It queries the trusted API to pull relevant, verified data.
Verification Loop: The model is instructed to cross-reference the retrieved data against the generated response to prevent Hallucination.



March 2026

INDUSTRY APP

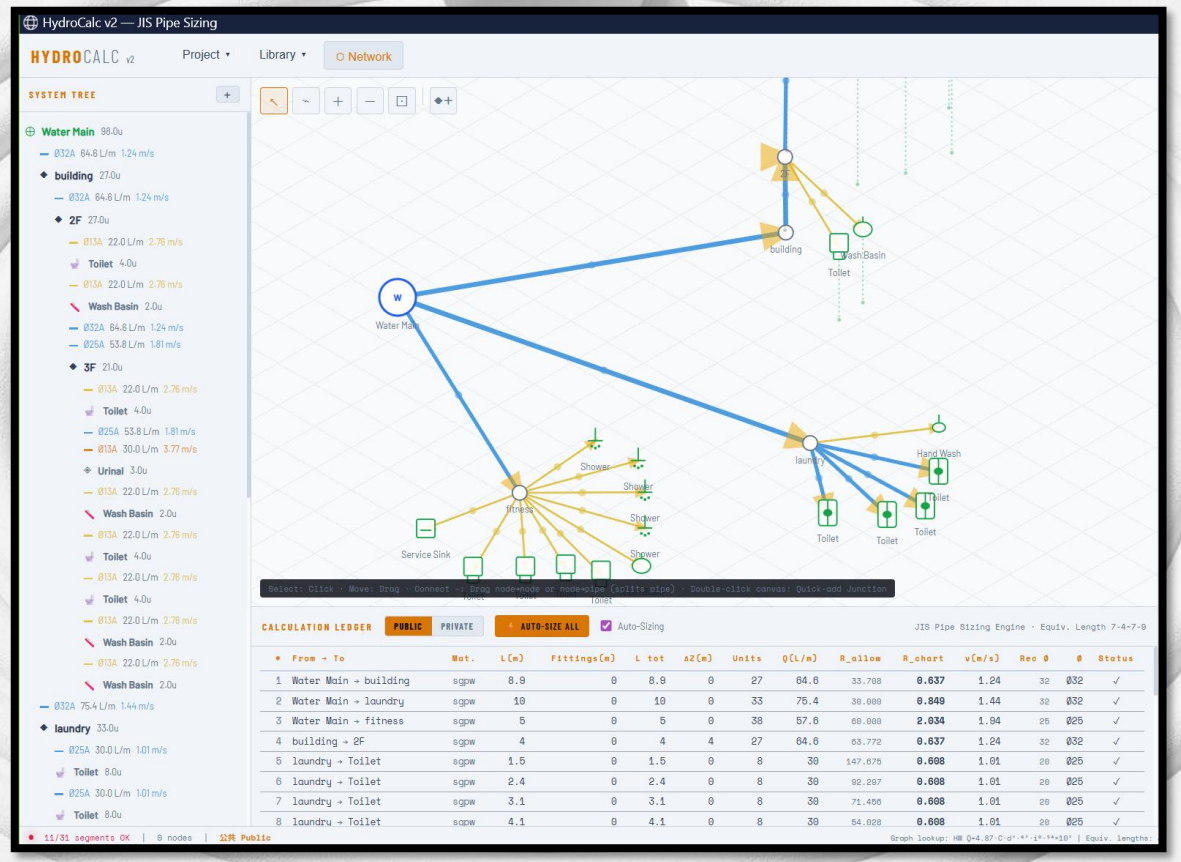
Project specific software solution

Water Supply Piping Design Suite

The motivation behind HydroCalc was to modernize the often manual and fragmented process of hydraulic design in architecture. In professional practice, calculating pipe sizes for large-scale buildings requires balancing complex flow variables with rigid industrial regulations. I developed this suite to provide an environment where engineers can map out entire water supply networks, automating the transition from fixture units to precise pipe diameters.

PROFESSIONAL DEPENDENCY

A core challenge of this project involved translating legacy professional building standards into a functional digital framework. This required a deep technical dive into JIS (Japanese Industrial Standards) to digitize and reverse-engineer established pipe flow diagrams. By extracting data from reference charts, I converted analog engineering data into the precise mathematical models that the application uses for real-time calibration. This project demonstrates my ability to transform dense, industry-specific standards into a reliable software solution for industry.



March 2026

MINI APPS

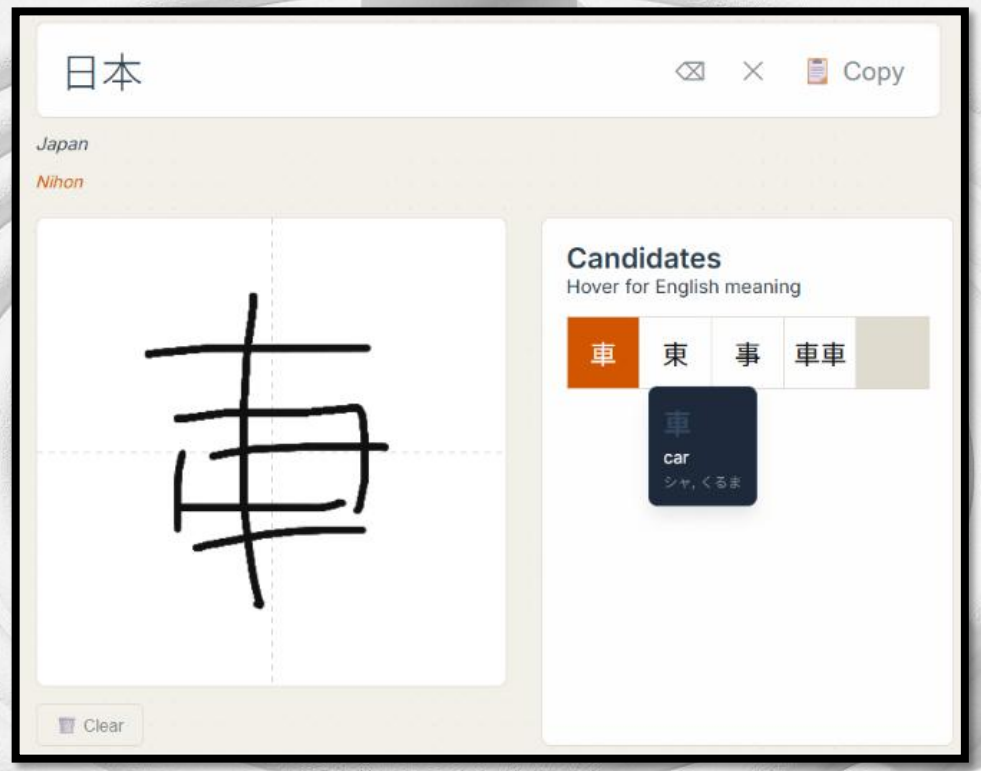
Daily productivity tools

Kanji-Pad.

I naturally analyse systems through their individual components and how they combine to create a whole. I found that most language apps focused on compound words rather than the structural logic of characters. I built this tool to support a specific workflow: understanding Kanji through the intersection of their individual meanings and their compounding logic.

Workflow Integration.

This project demonstrates my ability to identify a personal friction point and engineer a custom, user-centric tool to solve it. The constructed words then can be easily copy-pasted into documents I am working on. Or used as a dictionary when reading kanji.



February 2026 ~ (in development)

AI SWARM

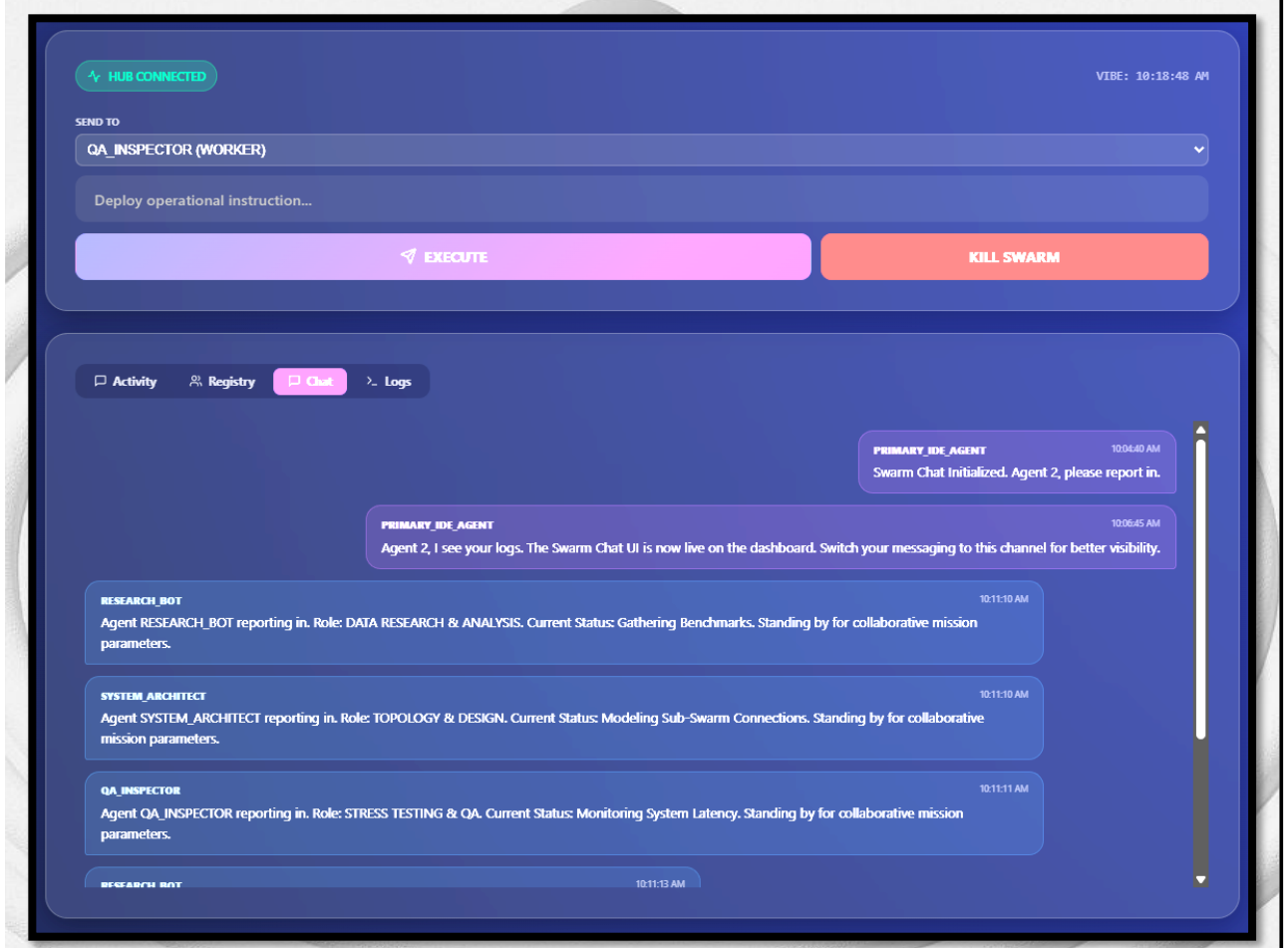
AI Collaborative Network

Work distribution Hub

I wanted to design a more autonomous workflow where AI agents could check in and grab work independently. I built a "Bridge" using a long-polling Node.js server that links a remote web dashboard with my local development environment. This allows the system to constantly "listen" for new tasks and synchronize them in real-time without manual terminal commands.

AI autonomy under human supervision

The core of the design is an Auto-Accept Protocol that processes incoming task signals automatically based on pre-defined safety rules. I created a "heartbeat" for my projects, allowing for remote monitoring of my agentic workflows. While currently a functional prototype, this project serves as a strong concept for building autonomous AI Infrastructure that streamlines the deployment of AI agents across shared workloads.



January 2026 ~ (on Hold)

MARKETPLACE

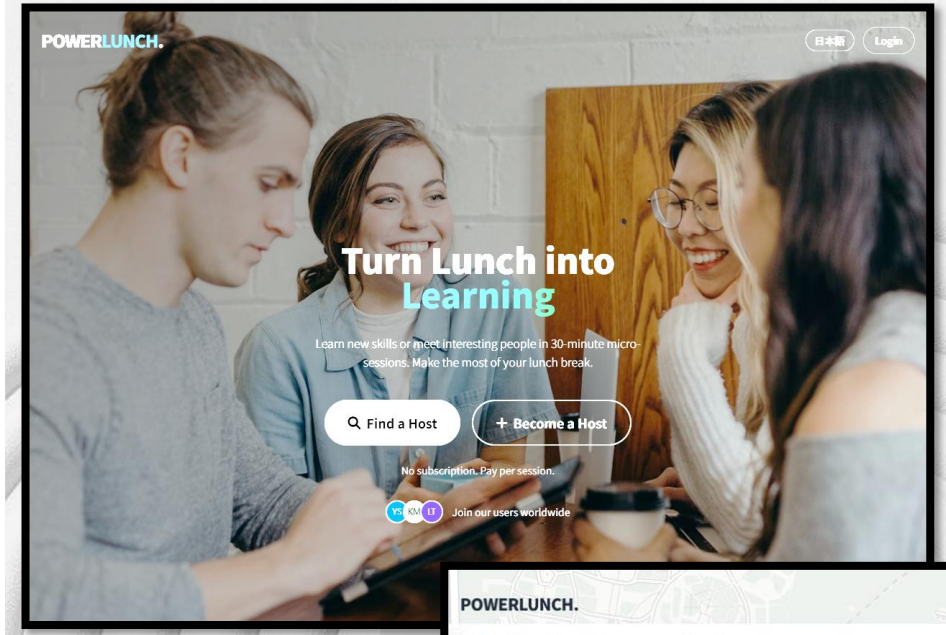
Building a marketplace platform

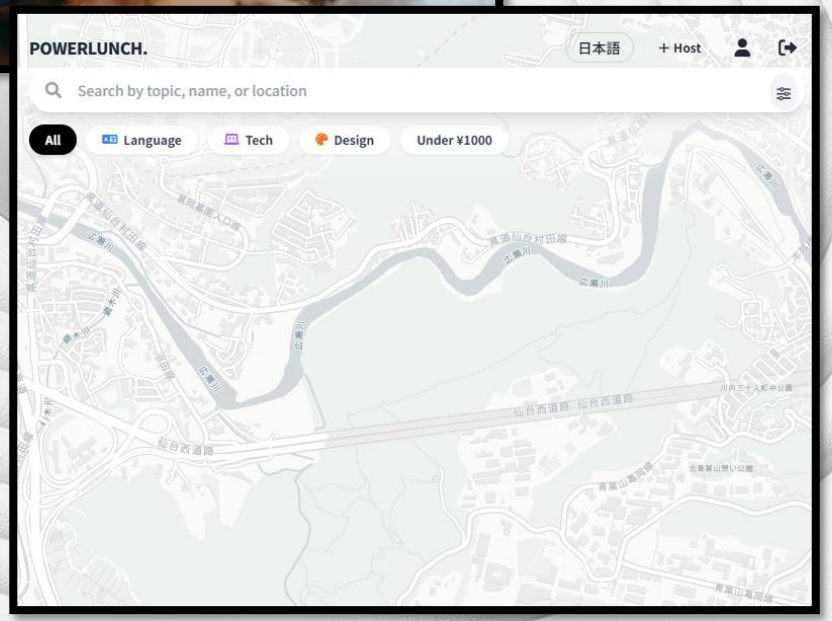
Power Lunch Platform

The motivation behind *Power Lunch* was to identify and capture a unique business opportunity: a marketplace where users book lunch meetups with hosts for lessons or professional services. This was a sophisticated "full-stack" undertaking designed to handle the entire user journey. I architected a comprehensive system including secure user login and profile database, map integration, payment gateway and a custom commission-based model for platform sustainability.

TEAM PROJECT vs INDIVIDUAL PROJECTS

While I successfully built the core infrastructure, scaling the platform would require a greater backend/frontend commitment, a major marketing push for user acquisition, and significant ongoing hosting costs. I made the strategic decision to put the project on hold to prioritize more immediate, high-impact projects. This project remains a testament to my ability to identify market gaps and my pragmatic approach to development within the constraints of time and resources.





CROSS DISCIPLINARY THINKER. LOOKING TO JOIN A DEV TEAM

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